

Milestone 1 - First Playable: Progress Document

Sentinel Protocol

Naz Yagmur | Individual Project | Unreal Engine 5.8 (Blueprint-only)

Project title	Sentinel Protocol
Project file	NAZ_3023.uproject (Unreal Engine 5.8)
Arena map to open	Content/SentinelProtocol/Maps/SentinelProtocol_MainMap
GitHub	https://github.com/nazyg/NAZ_3023
Demo video	https://youtu.be/Y8cukKvGRQs

1. Arena Layout and Player Route

- The player spawns in the **Entry area (grey)**, facing a staircase that goes up.
- Stairs up lead to the **Upper Balcony (purple light and material)**: railings along the edges, open sky for now (roof intentionally left open).
- The player comes back down and passes through the **sliding door** (BP_SlidingDoor) to the staircase that leads to the lower level.
- **Lower Level (teal)**: large hall with one separate **Control Room** lit in **red** in a corner.
- At the far corner (opposite the spawn corner) a staircase leads up to the **Main Floor (blue)**, an open-air area.
- Three height levels (lower, entry/main, upper) are connected by stairs; every zone has its own colour/light so the player always knows where they are.

2. Milestone 1 Requirements - Status

Requirement	Status
Complete arena blackout	All major areas built with placeholder geometry (floors, walls, stairs, platforms, railings): entry, upper balcony, lower level, control room, main floor.
Player scale and metrics	Walked with the Third Person character in PIE: stair steps, corridors and door opening feel correct.
Traversal	Full route (spawn - upper - door - lower level - control room area - main floor) walked end to end in PIE (screen-recorded). No dead ends or missing floors.
Routes and flow	Multi-level route returns the player to the main floor at the opposite corner from the spawn. More alternate routes/shortcuts to be added after playtesting.
Verticality	Three levels connected by stairs; balcony with railings gives a high-ground position over the entry.
Preliminary spawn planning	PlayerStart in the entry area. Four Target Points mark future enemy drone spawns: D1-D2 on the main floor, D3-D4 in the control room.
Collision and playability	Blockout meshes use default collision. Blocking Volumes added along the open-roof edges so the player cannot fall out of the arena. No stuck points found in testing.
Basic visual language	Colour + coloured lights per zone: grey (entry), purple (upper), teal (lower), red (control room), blue (main floor).
Initial interactive feature	Sliding door: Blueprint actor with a box trigger and a Timeline. Opens upward

	when the player enters the trigger, closes after a 1.5 s delay when they leave (see section 4, Figures 7-8).
Iteration from Checkpoint 1	Several changes made after testing the space in PIE (see section 5).
Project organization	Content/SentinelProtocol/ with Audio, Blueprints, Maps, Materials, Meshes, UI, VFX. BP_SlidingDoor is in Blueprints; single main map in Maps.

3. Screenshots (Play In Editor)



Figure 1 - Entry area (grey): spawn point and the first staircase.

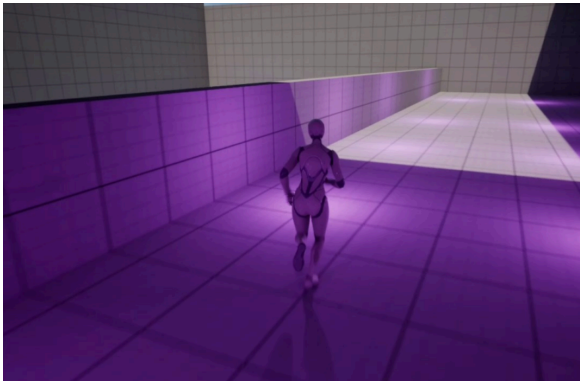


Figure 2 - Upper Balcony (purple): railings and open roof.



Figure 3 - Lower Level (teal): main hall.



Figure 4 - Control Room (red) entrance in the corner of the lower level.

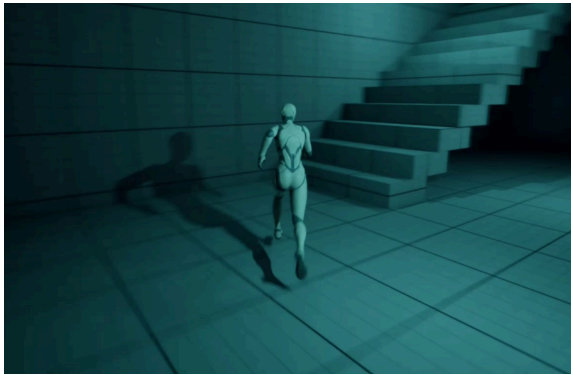


Figure 5 - Staircase from the lower level up to the main floor.

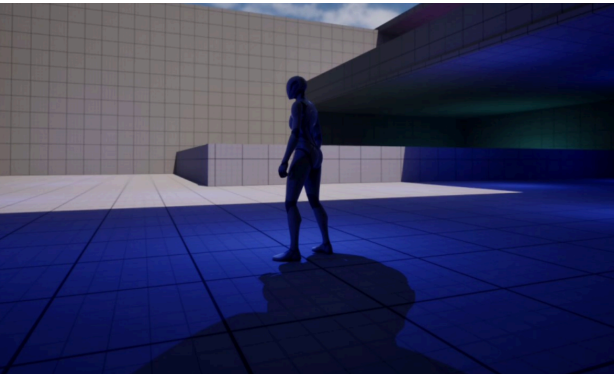


Figure 6 - Main Floor (blue): open-air area, arrival point.

4. Interactive Feature - Sliding Door (BP_SlidingDoor)

- Blueprint Actor with components: DoorMesh (static mesh cube scaled to the doorway), **TriggerZone** (box collision, overlap only).
- **DoorTimeline**: float track Alpha, 0.0 to 1.0 over 1 second.
- Two Vector variables, **ClosedLocation** (0,0,0) and **OpenLocation** (0,0,250). Timeline Update feeds **Lerp (Vector)** into **Set Relative Location** on DoorMesh, so the door slides up.

- **ActorBeginOverlap** calls Timeline Play (door opens); **ActorEndOverlap** waits a Delay of 1.5 s, then calls Reverse (door closes).
- Problem found in testing: the door closed on the player while walking through. Fixed by enlarging the trigger box along the walking direction and adding the 1.5 s delay before closing.
- Next: place the same Blueprint at the Control Room entrance as a chokepoint (not placed yet).

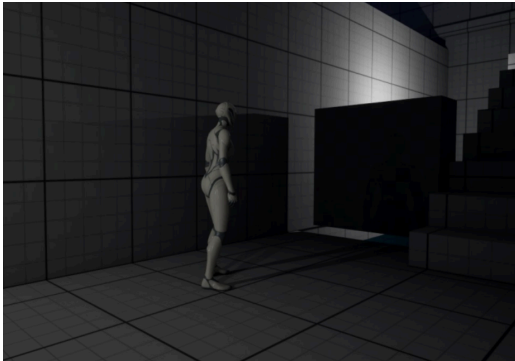


Figure 7 - Sliding door closed.

Figure 8 - Sliding door open (passage is clear).

5. Changes from Checkpoint 1

- Ramp replaced by stairs for moving between height levels - easier to build and read in the blockout; the stairs also tested well in PIE.
- Elevator replaced by stairs for the lower level to main floor connection. The elevator stays in the plan as an interactive feature for a later checkpoint.
- The Lower Tunnel became a larger lower hall so there is more room for future combat and drone spawns.
- Control Room moved to a corner of the lower level instead of being a separate final zone - keeps the arena compact and gives a clear red-lit destination.
- Roof left open on the main floor and balcony for now; Blocking Volumes keep the player inside.
- Control Room door not placed yet (the Blueprint is ready to reuse).

6. Known Issues and Next Steps

- Unreal shows a Virtual Shadow Map warning ("max lights overflow") where many local lights overlap. It does not affect gameplay; light count and overlap will be reduced during the polish pass.
- Some corridors on the lower level are dark - to be tuned with light placement after playtesting.
- Add the sliding door to the Control Room entrance.
- Add more alternate routes/shortcuts based on playtest notes.
- Checkpoint 2 goals: more interactive mechanics (elevator, hazards such as electrified floor panels) and the first combat/damage foundations.